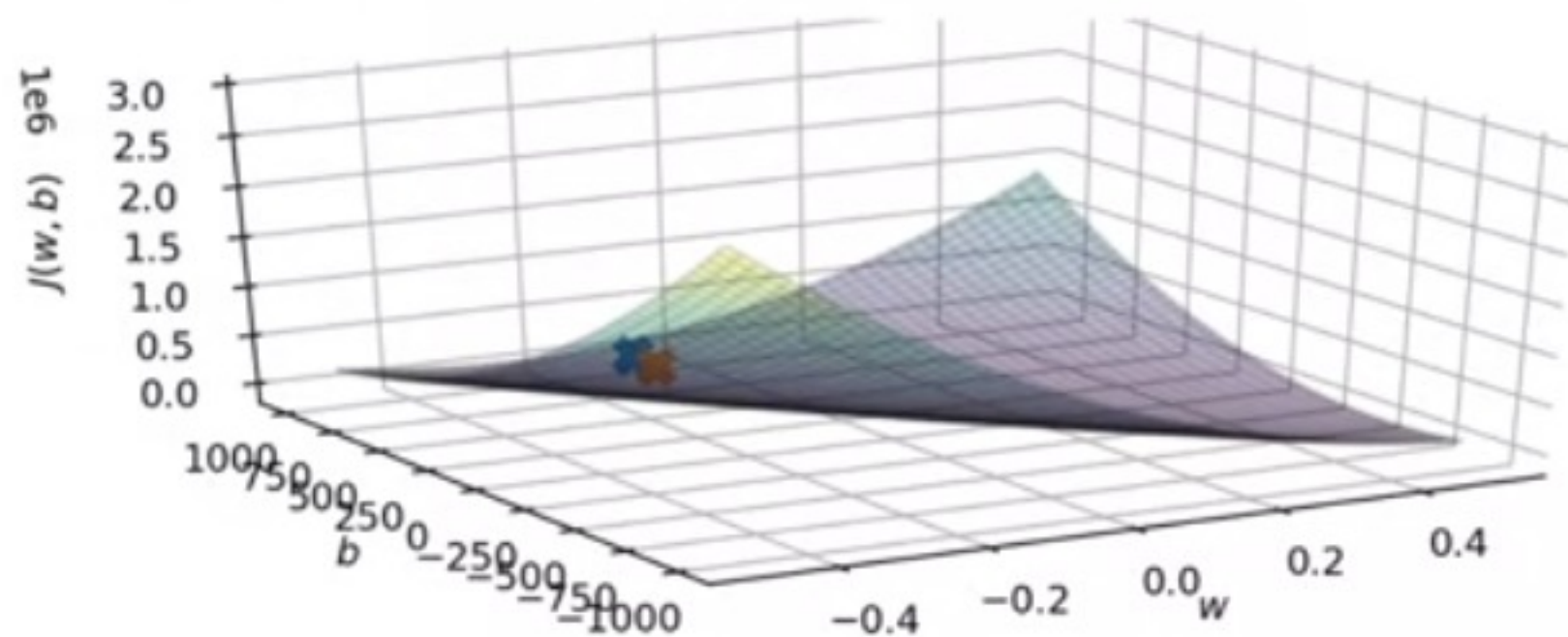
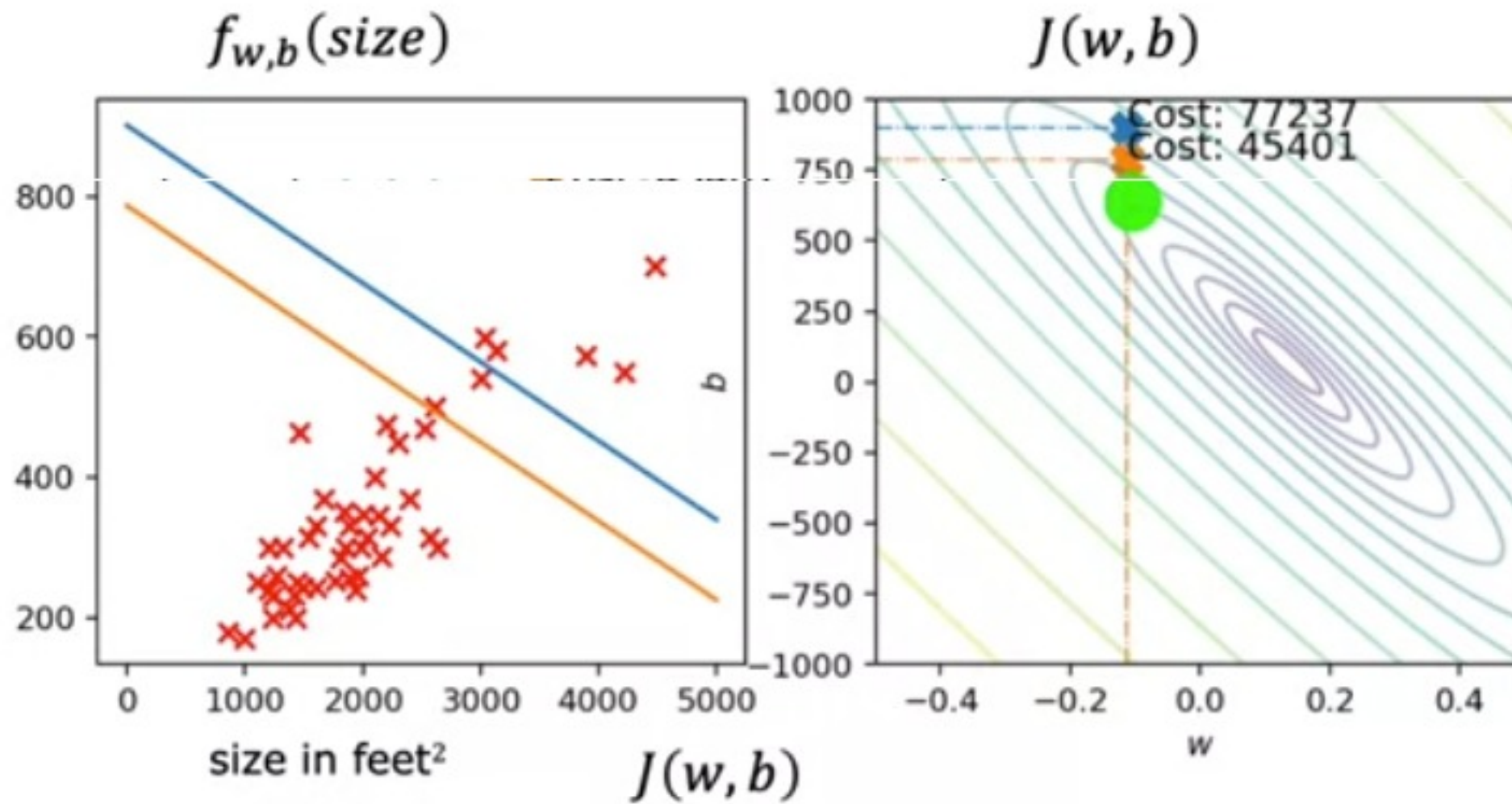


price in
\$1000's



here to this point just down and

Running gradient Descent

Let's see the algo in action:

Top left is the plot of model of data.

Top Right is a contour plot of the cost function

Bottom is surface plot of same cost function

Often, w & b will both initialize to 0

but for this demonstration:

Initialize $w = -0.1$ & $b = 900$

Blue to orange : and so on:

when you reached global minima

you could use this to predict now as in

image 1250\$. one.

"Batch" Gradient Descent:

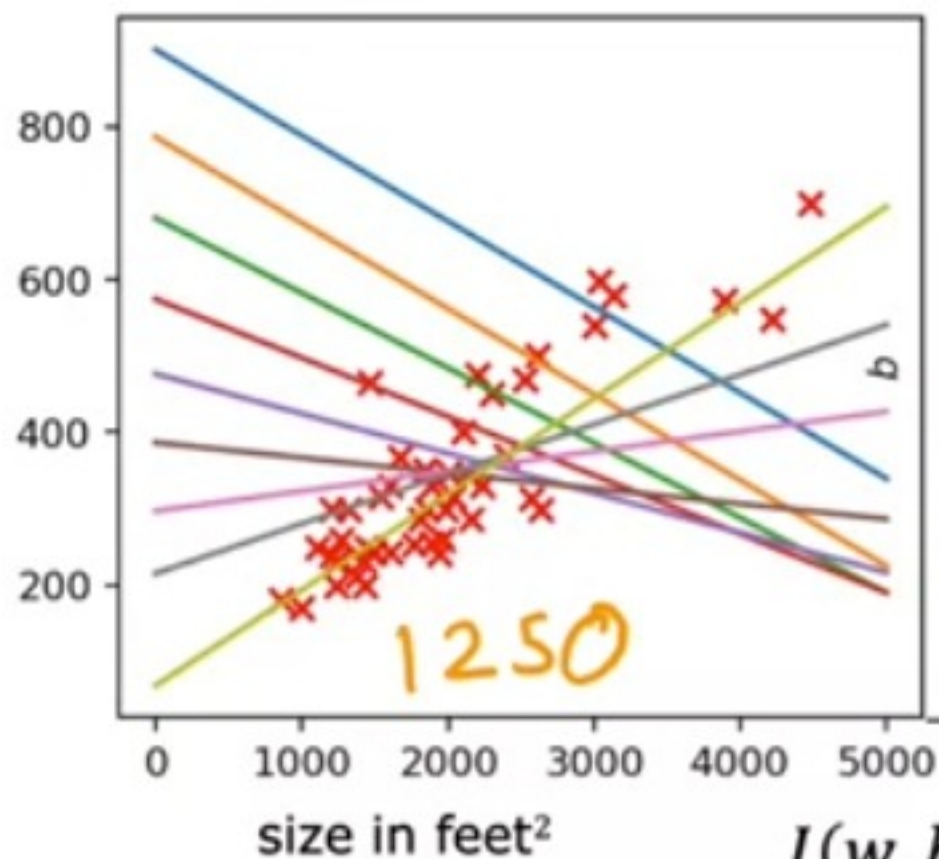
To be precise this process is also called

"Batch" Gradient Descent.

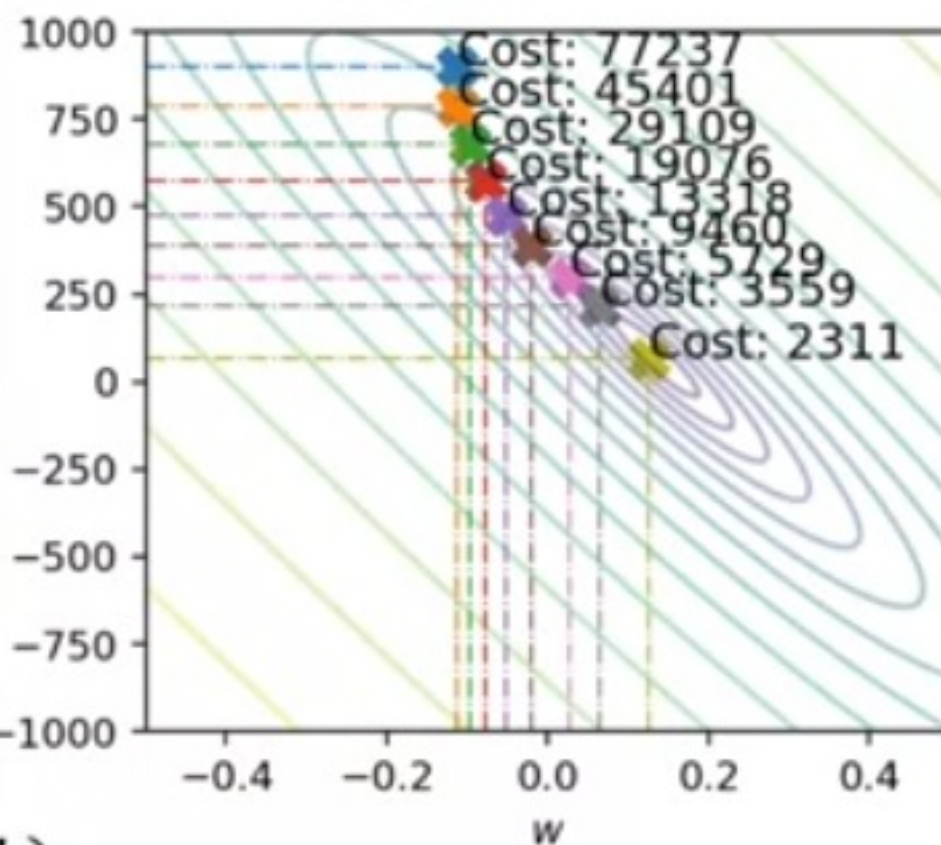
Batch : Each step of gradient descent uses all training examples.

price in
\$1000's

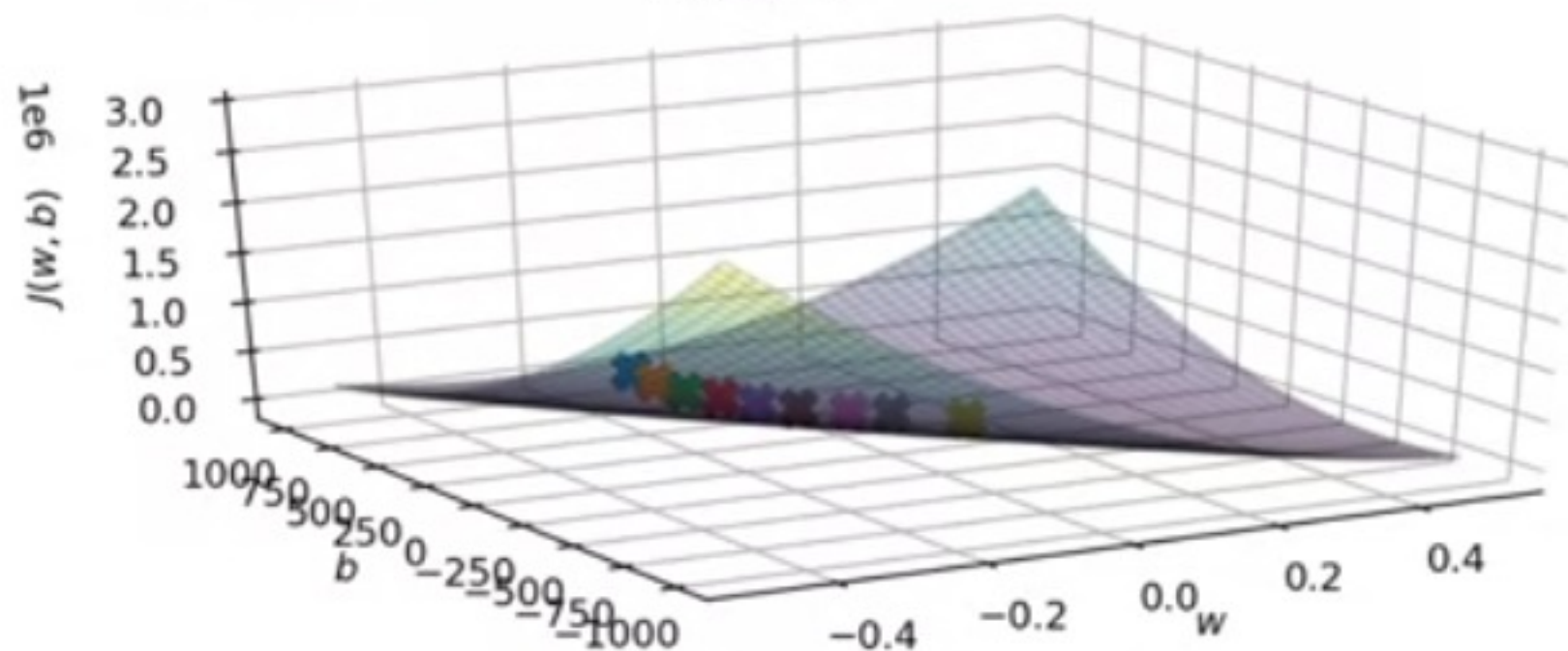
$$f_{w,b}(\text{size})$$



$$J(w, b)$$



$$J(w, b)$$



x size in ft^2	y Price in 1000's \$	$m=47$ L
1) 2104	400	$\sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$
2) 1416	232	
3) 1534	315	
4) 852	178	
47) 3210	870	

→ There are other gradient descent, that doesn't look at entire datasets.

price in
\$1000's

